

260816

Document Purpose

Note on Friday 14-Aug customer Review of Broke System export

Referring to the files in Salis's message

<https://ensembleanalyticsteam.slack.com/archives/C0B4D92FE30/p1786644000849329>

Loom

2 Loom to support this

Part 1 <https://www.loom.com/share/aff86a0925474a8b904a2f241deb334a>

Part 2 <https://www.loom.com/share/4dce4ad5a0564c119e2d0a607bd4d087>

Note.

You can search PDFs by text. This is really useful to be able to file icons in the pdf diagram

Abbreviations

Problem. Not all Functional Location Description words have been abbreviated when abbreviations already exist.

File: Broke System.functional_locations.xlsx

Column: Functional Location Description

Cell F15

Cell Value: 35-24L004 BROKE CONVEYOR 2

Feedback: CONVEYOR *must* be abbreviated to CVYR. There are several other examples too. Some words are already abbreviated but some are not.

Solution: You *must* use an abbreviation where there is one. Revisit the Abbreviation mapping logic. Use Regsex to extract every word in the Functional Location Description field, and abbreviate where an abbreviation already exists.

Empty Work Center column

File: Broke System.functional_locations.xlsx

Column: Work Center

Problem: the column is empty

Solution. Populate the column. Similar values to the equipment spreadsheet

Subordinate Equipment

File: Broke System.equipment

Column: Position

Problem: Superordinate Equipment Position must reset

See the screen shot below where I have corrected the position. See line 3 is subordinate equipment and its position must reset. And then line 4 continues from line 2. And line 5 is another subordinate equipment so its position resets.

Picture Subordinate equipment and another layer of hierarchy in the tree.

Solution.: Each new piece of subordinate equipment must reset the numbering

0	5001-PM03-BR-BR1-35-24L001	35-24LC-576		0040
1	5001-PM03-BR-BR1-35-24L001	35-24NC-577		0050
2	5001-PM03-BR-BR1-35-24L001	35-24-095		0060
3	5001-PM03-BR-BR1-35-24L001	35-24LV2-576	35-24-095	10
4	5001-PM03-BR-BR1-35-24L001	35-24-096		70
5	5001-PM03-BR-BR1-35-24L001	35-24-207	35-24-096	10
6	5001-PM03-BR-BR1-35-24L001	35-24-217	35-24-096	20
7	5001-PM03-BR-BR1-35-24L001	35-24-1105	35-24-096	30
8	5001-PM03-BR-BR1-35-24L001	35-24-093		80
9	5001-PM03-BR-BR1-35-24L001	35-24-234	35-24-093	10
0	5001-PM03-BR-BR1-35-24L001	35-24L001.1		90
1	5001-PM03-BR-BR1-35-24L001	35-24L001.2		100
2	5001-PM03-BR-BR1-35-24L002	35-24-002.1		0010
3	5001-PM03-BR-BR1-35-24L002	35-24XS-662		0020
4	5001-PM03-BR-BR1-35-24L002	35-24LC-622		0030
5	5001-PM03-BR-BR1-35-24L002	35-24GS-625		0040

Equipment Text – Line Identifiers

File: Broke System.equipment
Column: Equipment Text

Problem: All Line's Equipment text must start with LN.

Example 1

Cell F12 value is "35-24-095 PRESS PLPR PP-200 LINE".
But it should be "LN 35-24-095 PRESS PLPR PP-200"
The word "LINE" is removed but the text is prefixed with LN.

Example 2

Cell F14 value is "35-24-096 PRESS PLPR PP-900 LINE".
But it should be "LN 35-24-096 PRESS PLPR PP-900"
The word "LINE" is removed but the text is prefixed with LN.

Solution. Ensure all Line's Equipment Text are prefixed with LN and the word LINE does not appear in the text

Equipment Text – Valve Identifiers

File: Broke System.equipment
Column: Equipment Text

Problem: All Hand Valve Equipment text must start with HV

Example

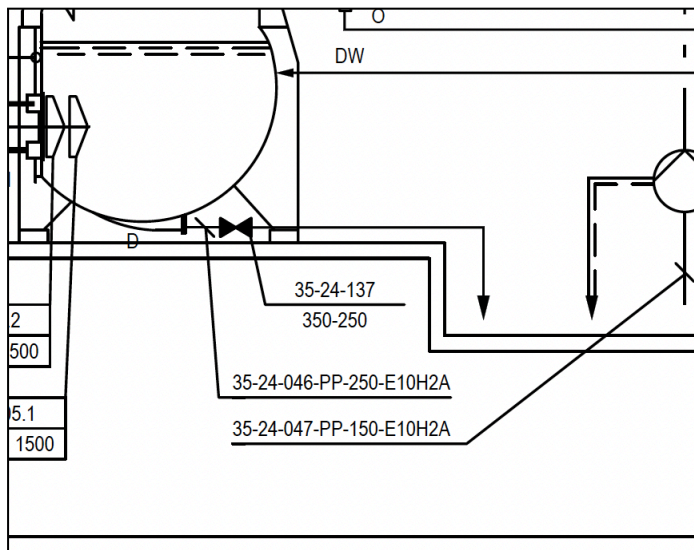
Cell F95 value is "35-24HV-548 HAND VLV"
But it should be "HV 35-24-548 35-24L005 AV".
Note that the middle HV has been removed from the text.
Where 35-24L005 = the REEL PLPR, refer to Broke System.functional_locations.xlsx, cell F15
And AV is Automatic Value

See the value in the screen print below, you see it has a circle above it. Look at the diagram key and you'll see that it is an automatic valve! We will need to use Computer vision to make this determination!

LEGENDS:		INSTRUMENTS		PIPELINES	
<u>SYMBOLS:</u>			CONTROL ROOM INSTRUMENT WITH DCS CONTROL		PLANT AREA CODE
	HAND VALVE		CONTROL ROOM INSTRUMENT WITH MCS CONTROL		SUB-PROCESS CODE
	VALVE NORMALLY CLOSED		QUALITY CONTROL SYSTEM (QCS)		POSITION NUMBER
	CHECK VALVE		CONTROL ROOM INSTRUMENT		FLOW MEDIA
	PRESSURE REDUCING VALVE		LOCAL CONTROL PANEL INSTRUMENT		DN SIZE
	SAFETY VALVE		LOCALLY MOUNTED INSTRUMENT		PIPE CLASS
	FLUSHING		PLC CONTROL		
	SAMPLING				
	DRAINAGE				
	AUTOMATIC VALVE				
	AUTOMATIC VALVE, ELECTRICAL MOTOR ACTUATOR				
	SIGHT GLASS				
	ORIFICE PLATE				
	STEAM TRAP				

And looking at the key you will see it is a Drainage Valve that is normally closed

Example of an 35-24-137 Drainage Valve Normally Closed

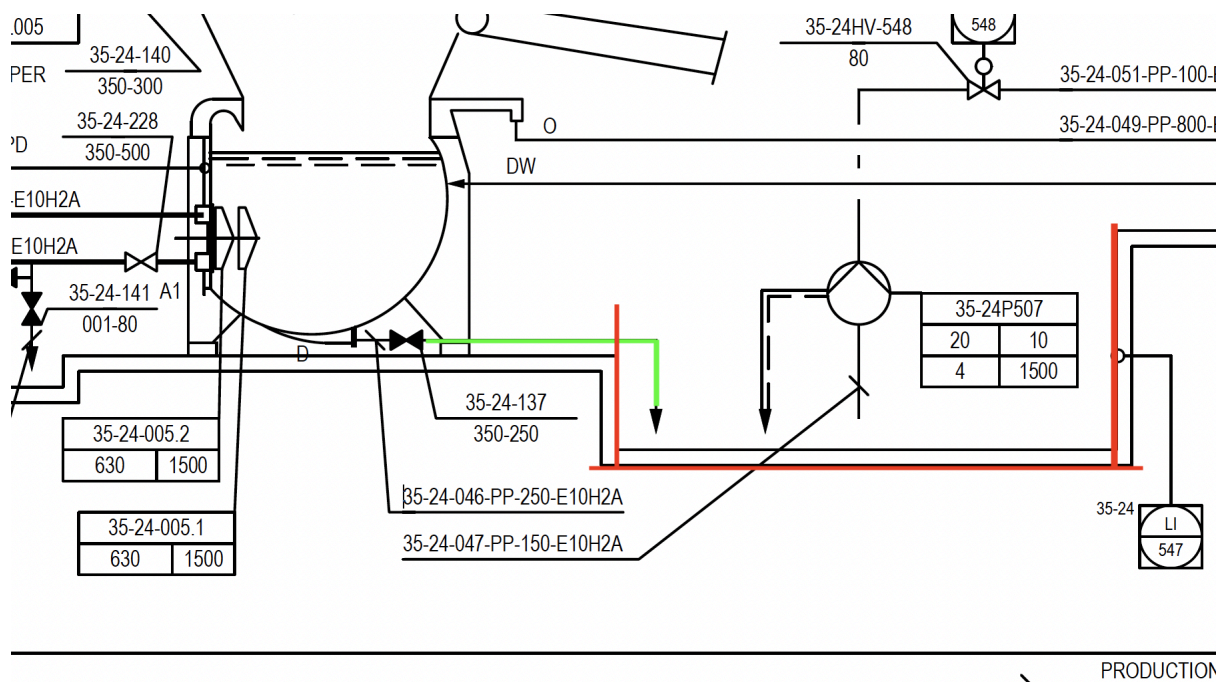
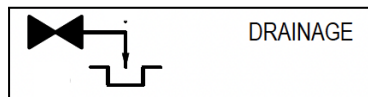


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Drainage Valve

We were wondering if the diagram was wrong. It is not. We just needed to zoom out.

Looking at the screen shot below, the green arrow is the arrow that shows the valve is a drainage valve. And the Red line marks out the drain



Naming Conventions

Conversation with Rob

Rich to Rob

Referring to the attached screenshot, on Friday we discussed the "Equipment Text" value for 35-24-137

We discussed that the Equipment Text value for this should be

HV 35-24-137 35-24L005 DRN NC

where 35-24L005 is the pointer to the Reel Pulper rather than using the text "Reel Pulper" or "35-24L005 Reel Pulper".

So we're saying that this format is *correct*

HV 35-24-137 35-24L005 DRN NC

and these two are *wrong*:

HV 35-24-137 Reel Pulper DRN NC

HV 35-24-137 35-24L005 Reel Pulper DRN NC

I *think* I see a lot of *inconsistency* with this. In the worked example "SML CHP1-Equipment Upload Template FINAL V2.4_Created.xlsx", it seems many valves use the Name rather than the Pointer. Here is a random selection from the worked example:

Please can you explain the difference.

I appreciate we might be missing something obvious as we don't seem to have the CHP1 pdf.

HV 0657 ROOF FAN 2 DMP 1&2 A
HV 0659 ROOF FAN 3 DMP 1&2 A
HV 0677 SMOKE ABATEMENT HATCH A
HV 0677 SMOKE ABATEMENT HATCH A
HV 42884 PI 1134 NO
HV 42254 SMPL STN NO
HV 42215 BLR WTR NO
HV 42255 BLR SFTY SHW NO
HV 42216 T.HALL BLW DWN TNK NO
HV 42236 AUX CNDT COOLG WTR NO
HV 42256 SFTY SHW DEMIN NO

Rob Response

Hi Richard,

There will be the human element here, where some have entered one way and some have entered another for CHP1. CHP1 will have errors due to the speed it needed to be uploaded.

I think that in an ideal world:-

HV 35-24-137 REEL PLPR 35-24L005 DRN NC

Would be best, but likely to be restricted by the character limit of 40 for some asset "names" rather than "numbers".

Referencing asset number only at least it has a reference point when looking at P&ID and when looking at tank number in a search it would bring up these assets too.

HV 35-24-137 35-24L005 DRN NC

I would say is a compromise but should be usable throughout the plant within the character limit for lines and valves. This indicates that it is the "drain valve for 35-24L005 that is normally closed".

This was outlined in our master data SOP.

Technical Object Description

All descriptions have a maximum character limit of 40 characters and they are to be all in uppercase text.

Where possible for valves we would like to capture the running condition, whether it is normally closed (NC) or normally open (NO). If the valve in the process is a process controlled valve we are not adding any further information.

They are to follow the following format as best as possible.

Asset Identifier-Description-Running Condition(if applicable)

Examples:-

42T6103 PRE-TREATED WTR TNK
LN 42-46627 42P5107 SUCT
HV 4211 42P5107 SUCT NO
HV 4212 42P5107 SUCT DRN NC
42P5107 PRE-TREATED WTR PMP 1

Refer to the SOP on Drive

https://docs.google.com/document/d/1KloDDtiqhoTieB_wDvt_30G608aHdegy/edit?usp=drive_link&oid=112243322026164044323&rtpof=true&sd=true